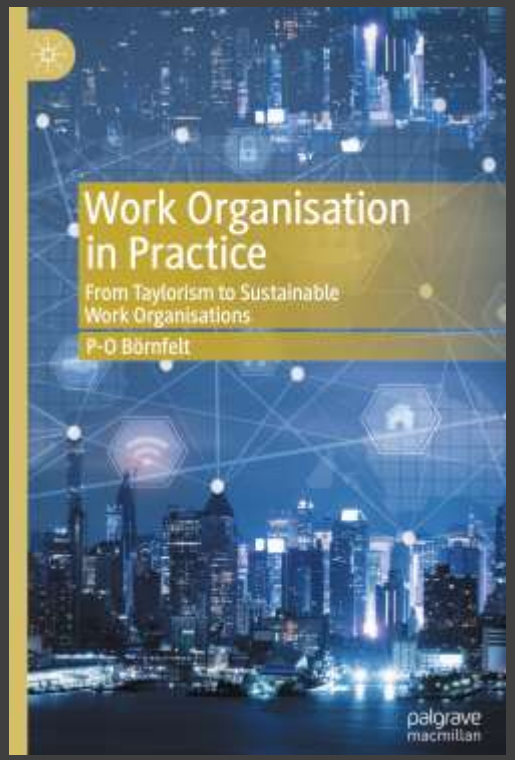
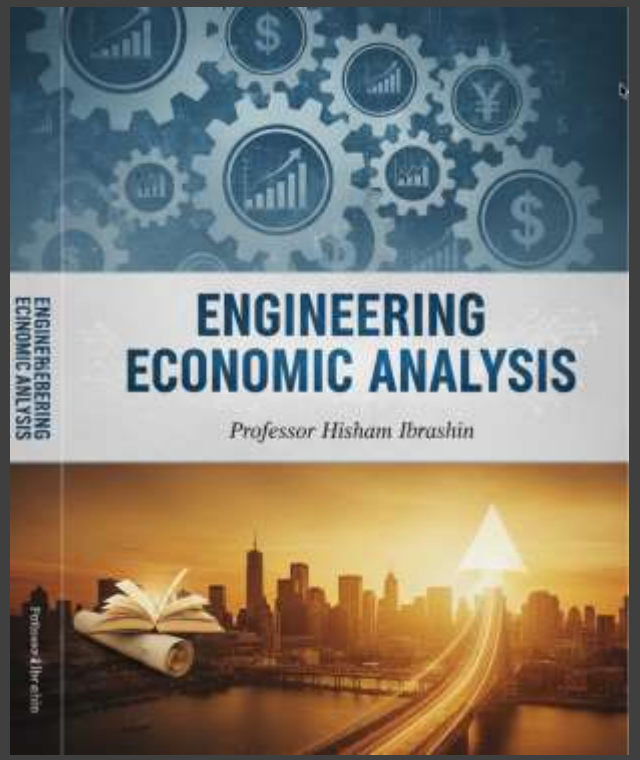
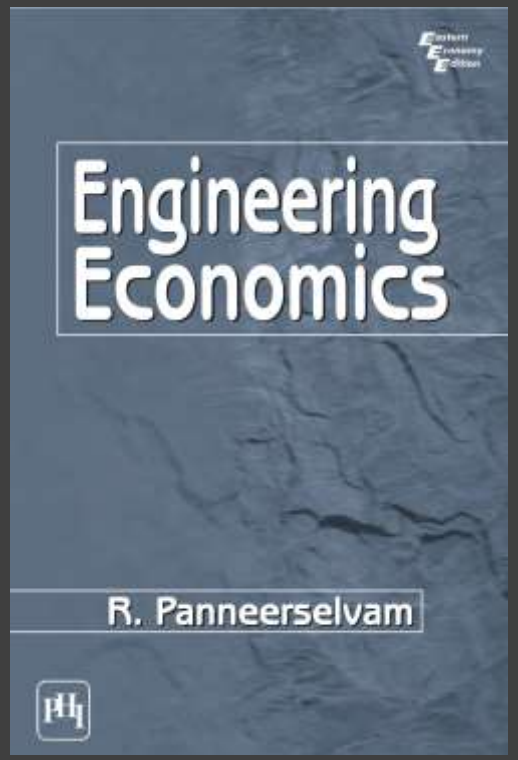
IND3001 Engineering Economics

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Course Material

This course asks two questions at once:

- **"Should this investment be made?"**
  The standard tools of engineering economics (the time value of money, cash flow analysis, NPV/IRR, depreciation, break-even, risk analysis) produce a numerical answer. Students learn to use these tools competently.
- **"Why is the question posed in this way, who is asking it, and for whom is the answer 'correct'?"**
  The same tools acquire their meaning inside a particular historical mode of production: within capitalist production relations and the logic of capital accumulation. Students learn which social relations the number they compute makes visible, and which it conceals.

3,000,000 TL — 4 workers — 6 years

- "There is a robot cell that can do the work of the four workers on the assembly line. It costs 3 million lira. It needs 300 thousand lira of maintenance a year. It lasts six years, and at the end we sell it as scrap for 300 thousand. The four workers cost us 600 thousand lira each per year — wages, social security, meals, shuttle. Should we buy it or not? I want one page by Thursday."

Central aim?

- The central claim of the course is this: **the discipline of industrial engineering arose out of the need to measure and control the labour process on behalf of capital** (Taylor, Gilbreth, Gantt).
- Engineering economics is that discipline's monetary answer to the question "which machine, which line, how many workers?" Knowing this historical origin does not make an engineer abandon the technique; it allows the engineer to see what the technique does.

Learning Outcomes /**Technical**

- Draw cash flow diagrams and perform equivalence calculations using the time value of money.
- Evaluate investment alternatives using NPV, IRR, payback period and annual worth, and interpret cases where the criteria disagree.
- Apply cost structures, depreciation and break-even analysis to a production or automation decision.
- Make decisions under uncertainty using sensitivity analysis and simple decision trees and show the effect of inflation on the calculations.

Learning Outcomes / **Critical**

- Explain the fundamental differences between neoclassical (marginalist) value theory and classical/Marxian labour theory of value and discuss how that difference is reflected in the concept of "cost."
- Evaluate an investment decision in terms of the labour process (control, skill, intensity), employment, and capital accumulation.
- Explain the historical origin of industrial engineering (scientific management) and the engineer's position within production relations (the "new middle class" debate).
- Critically assess the effects of automation, global value chains and ecological externalities on employment and production relations.

Core concepts

- Value, price, cost, cost types (fixed/variable, sunk, opportunity, marginal)
- Time value of money: simple/compound interest; P, F, i, n; present and future value
- Cash flow diagrams; equivalence; uniform series (A); factor notation; nominal vs effective interest
- Investment criteria: NPV, IRR, payback period, AW
- Depreciation
- Cost–volume–profit; break-even point
- Risk and uncertainty: sensitivity analysis
- Inflation: nominal vs real interest
- Public investment; benefit–cost analysis; the social discount rate
- Life-cycle cost; environmental externalities

- Engineering economics is **the method of comparing and selecting among engineering alternatives on the basis of their monetary consequences.**
- Should a bridge be steel or reinforced concrete; should a machine be replaced now or in three years; should a line be automated or stay manual; should a plant be built in Bursa or Tekirdağ?
- All of these questions have technical answers — strength, capacity, logistics. But the final decision is almost always made in money.
- Engineering economics is the technique of converting technical alternatives into the same currency and placing them side by side.

technical backbone

- **Time value of money** — why 100 lira today differs from 100 lira a year from now; interest, present value, future value.
- **Cash flow analysis** — showing an investment's inflows and outflows over the years on a diagram and reducing them to a single number.
- **Investment criteria** — Net Present Value, Internal Rate of Return, payback period: three languages for the "do / don't" decision.
- **Cost analysis and break-even** — fixed and variable costs; how many units before we turn a profit.
- **Depreciation and taxes** — spreading a machine's value across years and its effect on tax.
- **Risk and uncertainty** — what happens if the forecasts fail; sensitivity analysis, decision trees.
- **Inflation, public investment, life-cycle cost** — extensions of the calculation into the real world.

Technical textbooks

- Blank, L. & Tarquin, A. (2018). *Engineering Economy* (8th ed.). McGraw-Hill.
- Newnan, D., Eschenbach, T., & Lavelle, J. *Engineering Economic Analysis*. Oxford University Press

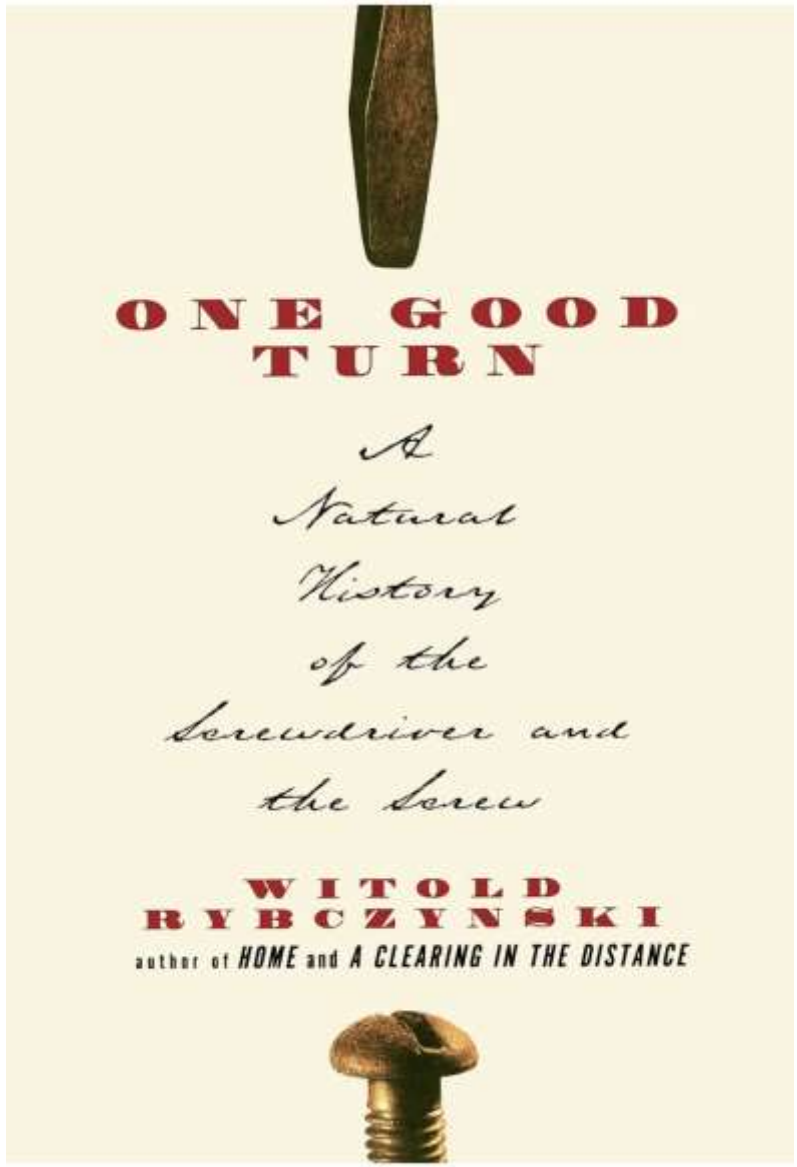
Critical and historical readings

- Thompson, P. & McHugh, D. (2003). *Work Organisations: A Critical Introduction* (3rd ed.). Palgrave. — Chapters on Taylorism, Fordism, the labour process, control, and post-Fordism.
- Marx, K. *Capital*, Vol. I: Ch. 1 (The Commodity), Ch. 15 (Machinery and Large-Scale Industry); Vol. III: the tendency of the rate of profit to fall (selections).
- Braverman, H. (1974). *Labor and Monopoly Capital*. — Ch. 4, "Scientific Management."
- Harvey, D. *The Limits to Capital | Seventeen Contradictions and the End of Capitalism*.

Historical and sociological readings

Bendix, Reinhard. *Work and Authority in Industry* — Managerial ideologies in the course of industrialisation.

- Burawoy, M. *The Politics of Production: Factory Regimes under Capitalism and Socialism* — The concept of the factory regime.
- Doray, B. *From Taylorism to Fordism: A Rational Madness* — The logic of Taylorism and its "rational madness."
- Börnfelt, P-O. (2023). *Work Organisation in Practice: From Taylorism to Sustainable Work Organisations*.
- Rybczynski, W. *One Good Turn: A Natural History of the Screwdriver and the Screw* — Standardisation, precision and industry through the history of an ordinary tool.
- Safley, T. M. (ed.) *Labor Before the Industrial Revolution: Work, Technology and their Ecologies in an Age of Early Capitalism*.
- Uchitelle, L. *Making It: Why Manufacturing Still Matters* — Deindustrialisation and the social meaning of manufacturing.
- Poulantzas / Ehrenreich — the "new middle class" and the engineer's class position.


ONE GOOD TURN
A Natural History of the Screwdriver and the Screw
WITOLD RYBCZYNSKI
author of HOME and A CLEARING IN THE DISTANCE


VİDA İLE TORNAVİDA
WITOLD RYBCZYNSKI